

Score-constrained ensembles and population transport

Status, 2026-10-01 (evening). Sections 1–5 (Sol/Astra) give the constrained action, compact-shape obstruction, positive non-Gaussian forward orbit and nonlinear-copy recoil/tail test. Sections 6–8 (Claude Fable) derive the two-sheet dynamical realization on any positive solution and on (18): balance, force, impulses, exact mean energy cancellation and all-time nonexplosion (Propositions 7–8), then prove that its controller is the field pair and the record history (Proposition 9, Corollary 10), pass the two-unread-width test with branch controllers and give a coordinate witness against a marginal-score controller (Proposition 11), and show closure at every $\kappa \geq 0$ (Proposition 12). Verdict (§8): the realization is a stochastic pilot-wave representation of the supplied law; the route from apparatus closure to positive action is closed, and the zero branch is exactly the autonomous branch. Physical constraint enforcement and positive action selection have not been derived.

Restricting the ordinary action of two classical phase sheets to the shared-score preparations gives the canonical Fisher field action exactly. The restriction is not preserved by free Newtonian transport. A separate population-transfer construction identifies its full transport-conjugate action; bridge phase alone omits reservoir costs. The resulting two-coordinate model satisfies continuity, but its shape family fails the full field equation, even after adding Gaussian width. An evolving Gaussian-polynomial construction instead gives an exact positive, non-Gaussian solution for every forward time. A nonlinear coordinate copy generates a shape outside every finite Hermite–Gaussian family, with an exactly retained recoil-energy cost.

These are variational identities, an explicit field construction and stated closure obstructions, developed by GPT-6.1 Sol and GPT-6 Astra on 2026-10-01. They do not derive a physical constraint mechanism, quantum recording, or positive action necessity. The supplied parameter is $\kappa \geq 0$; the field obstruction below concerns $\kappa > 0$. Proof status: written, internally checked.

1. What the classical sheet action supplies

Work on a connected position interval or the line. Require smooth positive densities, finite displayed energies, and vanishing boundary terms for the variations used below. A classical phase sheet has position density r and momentum W_q . Its canonical one-form and free Hamiltonian are $\int W \delta r dq$ and $\int r W_q^2 / (2m) dq$, respectively, with $m > 0$.

This one-form follows from the ordinary particle form. If material coordinates a carry fixed probability mass, $q = X(a)$, $r dq = da$, and $v = \delta X \circ X^{-1}$ is a virtual displacement, then $\delta r = -\partial_q(rv)$. Integration by parts gives

$$\int W_q(X(a)) \delta X(a) da = \int r W_q v dq = \int W \delta r dq. \quad (1)$$

Fixed total mass makes a spatially constant shift of W a gauge. This is a smooth variational identity; no infinite-dimensional well-posedness theorem is being assumed.

For two sheets of probability mass 1/2, introduce the constraint

$$r_\sigma = \rho/2, \quad W_\sigma = S + \kappa \sigma \log(\rho/\rho_{\text{ref}}), \quad \sigma \in \{+1, -1\}. \quad (2)$$

Here ρ is normalized, S has action units, and the positive constant ρ_{ref} only makes the logarithm dimensionless. Changing it adds sheetwise constants and changes no momentum. Formula (2) is the field version of the supplied positive classical preparation in the recording note, §8.22.

Proposition 1 (exact constrained action). Pulling the two-sheet one-form and energy

back to (2) gives

$$\begin{aligned}\Theta &= \sum_{\sigma} \int W_{\sigma} \delta r_{\sigma} dq = \int S \delta \rho dq, \\ \mathcal{H} &= \sum_{\sigma} \int \frac{r_{\sigma} W_{\sigma,q}^2}{2m} dq = \int \frac{\rho S_q^2}{2m} dq + \frac{\kappa^2}{2m} I(\rho), \\ I(\rho) &= \int \frac{\rho_q^2}{\rho} dq.\end{aligned}\tag{3}$$

The first identity cancels the two score signs; the second expands the two squares. Thus variation of the *restricted* action $\int [\int S \rho_t dq - \mathcal{H}] dt$ yields

$$\rho_t = -\partial_q(\rho S_q/m), \quad S_t = -\frac{S_q^2}{2m} + \frac{2\kappa^2}{m} \frac{(\sqrt{\rho})_{qq}}{\sqrt{\rho}}.\tag{4}$$

No further canonical-field bracket is needed after adopting the constraint and restricted variation. The Fisher modification is established prior art [HallReginatto2002; reading: passages pp. 3–8, especially the classical action (4) and modified action/equations (17)–(18); source companion]. The calculation here identifies which correlated sheet restriction produces that action and tests its invariance.

Why this is not an invariant reduction. Unrestricted classical sheet continuity gives, on (2),

$$\partial_t(r_+ - r_-) = -\frac{\kappa}{m} \partial_q^2 \rho.\tag{5}$$

For $\kappa > 0$, no strictly positive normalized density on the whole line has $\rho_{qq} \equiv 0$: an affine function cannot have those properties. Consequently every such constrained preparation departs from equal sheet densities immediately somewhere. Restricted variation and ordinary free sheet transport are different dynamics. Additional constraint reactions or state changes are required, as the exact Gaussian switching test in §8.23 also demonstrates. In particular (4)'s current uses the mean momentum S_q ; it does not give both physical sheet velocities $W_{\sigma,q}/m$ without further dynamics.

On connected positive support the labeled sheet momenta determine S_q and hence S up to a common constant. At exact vacuum, separate constants on disconnected components become invisible to the ordinary physical phase-space measure. This construction cannot carry them into a vacuum limit merely by declaring them canonical data.

2. Population transport includes the reservoirs

Let h_L, h_R be nonnegative smooth normalized bumps with compact, ordered, disjoint supports. Between them fix a bridge $B = [q_L, q_R]$. Choose a smooth positive normalized density ρ_0 with finite Fisher information. Put

$$\begin{aligned}h &= h_R - h_L, & \rho_w &= \rho_0 + wh, \\ G(q) &= \int_{-\infty}^q (h_L - h_R)(y) dy, & M(w) &= \int \frac{G(q)^2}{\rho_w(q)} dq, \\ R_B &= \int_B \frac{dq}{\rho_0(q)}.\end{aligned}\tag{6}$$

Here w is probability mass transferred from left to right, $0 \leq G \leq 1$, $G = 1$ on B , and G vanishes outside the reservoirs and their connector. If K is their compact enclosing interval, $c = \min_K \rho_0 > 0$ and $H = \|h\|_{\infty} > 0$, then $|w| < c/(2H)$ ensures $\rho_w \geq c/2$ on K . Thus all terms in (6) are finite, with $M(w) \leq 2|K|/c$, and smooth in this w range.

Vanishing exterior flux and continuity uniquely require $j(q) = \dot{w} G(q)$, because $\partial_q j = -\partial_t \rho_w$. The corresponding total transport kinetic energy is $mM(w)\dot{w}^2/2$. Both M and R_B have length-squared units. Smooth reservoir transport contributes positively, so $M(w) > R_B$.

Proposition 2 (continuity-compatible canonical family). Define an action field, up to a common constant, by

$$S_q(q; w, P) = \frac{PG(q)}{M(w)\rho_w(q)}. \quad (7)$$

The variable P has action units; it is not a point-particle momentum. Then

$$\begin{aligned} \Theta &= \int S \delta \rho_w dq = P dw, \\ H(w, P) &= \frac{P^2}{2mM(w)} + \frac{\kappa^2}{2m} I(\rho_w), \\ \dot{w} &= \frac{P}{mM(w)}, \quad \dot{P} = \frac{P^2 M'(w)}{2mM(w)^2} - \frac{\kappa^2}{2m} \frac{dI(\rho_w)}{dw}. \end{aligned} \quad (8)$$

These local reduced Hamilton equations preserve H while w remains in the positive-density range and satisfy the ambient continuity equation exactly.

Proof. Since $h = -G_q$, integration by parts yields $\int S h dq = \int S_q G dq = P$. The phase energy in (3) is $P^2/(2mM)$, proving the Hamiltonian and its equations. Their first equation gives $\rho_w S_q/m = \dot{w}G$, the required current. $\square$

In this family the actual bridge action difference is

$$\Phi_B = S(q_R) - S(q_L) = \frac{R_B}{M(w)} P. \quad (9)$$

Thus Φ_B generally differs from the population-conjugate action $P = \int S h_R dq - \int S h_L dq$. The latter is a reservoir-weighted phase-action difference. Replacing it by bridge endpoint phases omits the reservoir motion. In coordinates (w, Φ_B) the canonical two-form has coefficient $M(w)/R_B$, rather than one.

For a fixed bridge density the exact dual identities are

$$E_{\text{transport}, B} = \frac{mJ^2 R_B}{2}, \quad \inf E_{\text{phase}, B} = \frac{\Phi_B^2}{2mR_B}, \quad J = \frac{\Phi_B}{mR_B}. \quad (10)$$

The phase infimum is the bridge lemma, (131), written with S instead of $\hbar\theta$; it is attained in the stated weighted H^1 class. The last identity makes the two energies equal. A low-density connector makes a fixed phase difference cheap and a fixed population flux expensive. It selects no absolute action scale.

The costs add over arbitrary spatial cuts, because both integrals of $1/\rho$ and of G^2/ρ add. This is an exact spatial transport identity, not Newton's temporal cell action. In the positive-tail family of recording (133), R_B grows as ϵ^{-2} . If reservoir costs additionally obey $M - R_B \leq C$ uniformly, then $|P - \Phi_B| \leq |P|C/R_B$ and the bridge becomes asymptotically dominant. That extra bound must be checked; exact vacuum is not such a check.

3. Continuity does not close the full field equation

Equations (8) are exact for the restricted model. They do not imply that its fixed density family is invariant under (4)'s Hamilton–Jacobi equation. For a Gaussian base $\rho_A = N(0, A)$ and initial $w = P = 0$, the full equation gives

$$\partial_t S_q = \frac{\kappa^2 q}{mA^2}. \quad (11)$$

The family (7) instead permits only a multiple of G/ρ_A at that initial time. It vanishes outside a compact interval, whereas (11) does not. This is an immediate nonzero transverse term for $\kappa > 0$. It cannot be removed by a spatially constant action gauge.

Adding Gaussian width supplies the missing global quadratic phase at $w = 0$. The next calculation tests whether that repairs closure for nonzero population transfer.

Proposition 3 (a fixed compact transfer shape cannot close). Fix $m, \kappa, A > 0$ and a nonzero $h \in C_c^\infty(\mathbb{R})$ with $\int h = 0$. The density family $\rho_{A,w} = \rho_A + wh$, allowing width variation and sufficiently small positive and negative w , cannot be locally invariant under (4) while including zero-phase-gradient states for every such w , satisfying continuity with vanishing exterior flux, and having twice differentiable parameter trajectories. The same obstruction persists with Gaussian mean motion allowed. The claim concerns an open family of preparations, not all finite-dimensional models.

Proof. Set $r = \rho_A$ and $G = -\int_{-\infty}^q h$, so G is smooth and compactly supported. The integrated density tangents for width and transfer are $G_A = qr/(2A)$ and G . At an initially zero phase gradient, continuity first gives $\dot{A} = \dot{w} = 0$. Differentiating it and using (4) requires

$$\begin{aligned} \rho_w F_w &= c_A(w)G_A + c_w(w)G, \\ F_w &= \frac{2\kappa^2}{m} \partial_q \frac{(\sqrt{\rho_w})_{qq}}{\sqrt{\rho_w}}, \quad c_A = m\ddot{A}, \quad c_w = m\ddot{w}. \end{aligned} \quad (12)$$

Outside the compact supports, $\rho_w = r$ and $F_w = \kappa^2 q/(mA^2)$; hence $c_A(w) = 2\kappa^2/(mA)$ for every small w . At $w = 0$ this already matches the full expression, giving $c_w(0) = 0$. Evaluation at any point with $G \neq 0$ makes $c_w(w)$ smooth, since the left side is smooth while $\rho_w > 0$.

Put $u = h/r$. Differentiating the square-root ratio gives $\delta[(\sqrt{\rho})_{qq}/\sqrt{\rho}] = (u'' - qu'/A)/2$. Retaining the variation of the density multiplier in (12), the necessary linearized identity is

$$u''' - \frac{q}{A}u'' - \frac{u'}{A} + \frac{q}{A^2}u = cg, \quad g = G/r, \quad c = mc'_w(0)/\kappa^2. \quad (13)$$

Since $G' = -h$, one has $u = qg/A - g'$. Substitution yields

$$\begin{aligned} -g'''' + \frac{2q}{A}g''' + \left(\frac{4}{A} - \frac{q^2}{A^2}\right)g'' - \frac{4q}{A^2}g' \\ + \left(\frac{q^2}{A^3} - \frac{1}{A^2}\right)g = cg. \end{aligned} \quad (14)$$

The function g is smooth and compactly supported because $r > 0$. All four initial data vanish at a point outside its support. Uniqueness for this regular fourth-order linear ODE forces $g \equiv 0$, hence $h = -G' \equiv 0$, a contradiction. Allowing the Gaussian mean adds a term proportional to r in (12); tail matching forces its coefficient to zero and leaves the same contradiction. $\square$

The zero-phase-gradient and continuity assumptions are essential to the initial tangency test. The theorem permits isolated states, shapes depending on parameters or time, and other finite-dimensional families. It specifically rules out treating a fixed compact transfer bump plus Gaussian width as an exact free Fisher evolution. Generated shape information cannot be hidden inside that projection.

4. An exact positive nonlinear shape evolution

The compact-shape obstruction permits evolving tails. The following construction solves (4) exactly within its supplied dynamics. It also identifies a limitation of the first mode test: the linearized mode is a width/phase tangent, although its finite-amplitude completion is not Gaussian.

For $\kappa > 0$, introduce the auxiliary algebraic variable

$$\psi = \sqrt{\rho} e^{iS/(2\kappa)}, \quad \psi_t = \frac{i\kappa}{m} \psi_{qq}. \quad (15)$$

The second equation is equivalent to (4) on positive density: its real part gives $\rho_t = -\partial_q(\rho S_q/m)$ and its imaginary part gives (4)'s second equation after differentiating the displayed product. It

is a change of variables in the adopted field equations, not an independent physical propagation premise.

Fix $A > 0$ with length-squared units and put

$$\begin{aligned} u &= \frac{\kappa t}{mA}, & B &= A(1 + u^2), & s &= \arctan u, \quad z = q/\sqrt{B}, \\ \psi &= B^{-1/4} e^{iuq^2/(4B)} \chi(z, s), & i\chi_s &= \left(-\partial_z^2 + \frac{z^2}{4} \right) \chi. \end{aligned} \quad (16)$$

For completeness, $a = \dot{B}/(2B) = \kappa u/(mB)$ and $b = \dot{s} = \kappa/(mB)$ obey $\dot{a} + a^2 = b^2$. Differentiating the second line cancels the terms $-a\chi/2$ and $-az\chi_z$ on both sides of (15). The remaining quadratic phase coefficient is $bz^2/4$, giving the last equation in (16). Thus no propagator formula is assumed.

Let

$$\begin{aligned} \phi_0(z) &= (2\pi)^{-1/4} e^{-z^2/4}, & H_2(z) &= (z^2 - 1)/\sqrt{2}, \quad \phi_2 = H_2\phi_0, \\ \psi_n(q, t) &= B^{-1/4} e^{iuq^2/(4B)} e^{-i(n+1/2)s} \phi_n(z), & n &\in \{0, 2\}. \end{aligned} \quad (17)$$

Direct differentiation gives the oscillator eigenvalues $1/2$ and $5/2$ for ϕ_0 and ϕ_2 . Gaussian moments give unit norms and orthogonality. Hence both displayed functions solve (15).

Proposition 4 (positive forward completion). For $0 < \epsilon < \sqrt{2}$, the normalized combination $\psi_\epsilon = (\psi_0 + i\epsilon\psi_2)/\sqrt{1 + \epsilon^2}$ gives a smooth solution of (4) on the whole line for every $t \geq 0$:

$$\begin{aligned} \rho_\epsilon(q, t) &= \rho_G(q, t) \frac{1 + 2\epsilon \sin(2s)H_2(z) + \epsilon^2 H_2(z)^2}{1 + \epsilon^2}, & \rho_G &= N(0, B), \\ S_\epsilon(q, t) &= \kappa \left(\frac{uq^2}{2B} - s \right) + 2\kappa \arctan \frac{\epsilon \cos(2s)H_2(z)}{1 + \epsilon \sin(2s)H_2(z)}, \\ \frac{\rho_\epsilon}{\rho_G} &\geq \frac{(1 - \epsilon/\sqrt{2})^2}{1 + \epsilon^2} > 0. \end{aligned} \quad (18)$$

The lower bound is relative to the Gaussian envelope; it supplies no absolute density floor at infinity. All displayed energies and moments are finite. The density is non-Gaussian for every $\epsilon > 0$.

Proof. Factoring out ψ_0 leaves $F = 1 + \epsilon[\sin(2s) + i\cos(2s)]H_2$. For forward time, $0 \leq s < \pi/2$, so $\sin(2s) \geq 0$; $H_2 \geq -1/\sqrt{2}$ gives $\Re F \geq 1 - \epsilon/\sqrt{2} > 0$. Thus the arctangent in (18) is one globally smooth phase, and $|F|^2$ gives its density and bound. Orthogonality gives exact normalization. Gaussian decay times a polynomial, with this strictly positive denominator, justifies the derivatives, moments and integrations by parts. The nonconstant quartic factor cannot be absorbed into a different Gaussian width. Equations (15)–(17) prove the field equations. $\square$

The conserved energy and first two even moments are explicitly

$$\begin{aligned} E &= \frac{\kappa^2(1 + 5\epsilon^2)}{2mA(1 + \epsilon^2)}, & EQ &= 0, \\ EQ^2 &= B \frac{1 + 2\sqrt{2}\epsilon \sin(2s) + 5\epsilon^2}{1 + \epsilon^2}, \\ EQ^4 &= B^2 \frac{3 + 12\sqrt{2}\epsilon \sin(2s) + 39\epsilon^2}{1 + \epsilon^2}. \end{aligned} \quad (19)$$

Indeed (3)'s energy equals $2\kappa^2 \int |\psi_q|^2/m$. Integration by parts in (15) makes its time derivative zero. At time zero the imaginary coefficient cancels the derivative cross term, and $\int |\phi'_0|^2 = 1/4$, $\int |\phi'_2|^2 = 5/4$, giving E . For the moments use $E_0 z^2 = 1$, $E_0 z^2 H_2 = \sqrt{2}$, $E_0 z^2 H_2^2 = 5$, and $E_0 z^4 = 3$, $E_0 z^4 H_2 = 6\sqrt{2}$, $E_0 z^4 H_2^2 = 39$. Here E_0 integrates against ϕ_0^2 . The variance has

second derivative $4E/m$, as required by the free field virial identity; the linear-in-time term records the initial position/current correlation.

At first order in ϵ the paired tangents are

$$\delta\rho = 2\rho_G \sin(2s)H_2, \quad \delta S = 2\kappa \cos(2s)H_2. \quad (20)$$

They are width and quadratic-phase tangents, not an independent linear population-transfer mode. The result is their exact non-Gaussian nonlinear completion. It carries evolving tails rather than a fixed compact transfer bump. Restricting this one solution to any time partition is exactly consistent; no general continuum existence theorem or separated-packet apparatus has been constructed.

Forward time matters: at $s = -\pi/4$ the factor is $1 - \epsilon H_2$ and has real zeros. Thus (18) does not define a globally positive real-field family under all backward evolution. For fixed t , $\kappa \downarrow 0$ gives a stationary non-Gaussian density, $S \rightarrow 0$, and $E \rightarrow 0$. The construction leaves the zero branch admissible.

5. Recording generates a shape outside the finite mode family

Free completion must survive an actual intervention. Couple a body state (ρ, S) from (18), or any positive state with the following finite integrals, to a pointer configuration $Z \sim N(0, C)$, $C > 0$, with zero pointer action. Prepare the joint momenta with the shared score sign (2), so the full phase-space law need not factor. Apply the canonical copy (137) in the recording note, with smooth length-valued $f(q)$. In the algebraic variables its pullback is exactly

$$\Psi'(q, z) = \psi(q)\eta_C(z - f(q)), \quad \eta_C(y) = (2\pi C)^{-1/4}e^{-y^2/(4C)}. \quad (21)$$

The complete pointer-coordinate record $R = r$ gives

$$\rho_r(q) = \frac{\rho(q)e^{-(r-f(q))^2/(2C)}}{\int \rho(y)e^{-(r-f(y))^2/(2C)}dy}, \quad S_r(q) = S(q), \quad (22)$$

up to a record-dependent constant action. These are the exact conditional fields; the sign stays fair and independent of coordinates.

Proposition 5 (generated shape and retained recoil). For $f(q) = q^2/\ell$ with fixed length $\ell > 0$, every posterior (22) from (18) lies outside every finite Hermite–Gaussian expansion, even allowing a different Gaussian mean, width and quadratic phase. The mean body-energy increase is nevertheless exact:

$$\begin{aligned} \Delta E_{\text{body}} &= \frac{\kappa^2}{2mC} E_\rho[f'(Q)^2] = \frac{2\kappa^2}{mC\ell^2} E_\rho Q^2, \\ E_R I(\rho_R) - I(\rho) &= \frac{1}{C} E_\rho[f'(Q)^2]. \end{aligned} \quad (23)$$

All these posterior densities are smooth and positive, with finite Fisher information and body energy. The finite-mode failure is a failure of shape closure, not a violation of the conditional covariance bound (138).

Proof. The logarithmic posterior tail obeys $\lim_{|q| \rightarrow \infty} q^{-4} \log \rho_r(q) = -1/(2C\ell^2)$. A nonzero finite Hermite expansion over any Gaussian envelope has polynomial-times-Gaussian density and the same limit is zero. This proves the stated closure failure, without ruling out other finite models. For the energy differentiate (21): $\partial_q \Psi' = \psi_q \eta_C - f' \psi \eta'_C$. Integration in z kills the cross term since $\int \eta_C \eta'_C = 0$, and $\int |\eta'_C|^2 = 1/(4C)$. Multiplication by $2\kappa^2/m$ gives (23)'s first line, which is also the actual canonical recoil energy under (2). For its second line, the conditional score is $\partial_q \log \rho + (r - f)f'/C$; conditional on q , $r - f$ has mean zero and variance C . Expanding its square proves the identity. Quartic decay proves the stated posterior integrability. $\square$

If the coordinate record becomes unread, the marginal position density and mean current return to the original ρ and $\rho S_q/m$. The actual body energy keeps the increment (23). Resetting its momenta to that marginal density's two score sheets would erase precisely this energy. Here every conditional phase gradient is the same, so there is no additional current-variance excess; for general branches both excesses in recording (135) must be retained. The pointer energy is unchanged by this impulsive shear. The extra body energy therefore requires work from its actuator; a closed energy-supplying apparatus has not been derived. Full conjugate access remains the distinct terminal escape (140). None of these formulas selects $\kappa > 0$.

6. The two-sheet realization on any positive solution

The handout's candidate balance equations are now derived and checked on every smooth positive solution of (4), with the explicit orbit (18) as the test case. From here on $v = S_q/m$ is the mean velocity and

$$d = \frac{\kappa}{m} \partial_q \log \rho, \quad p_\sigma = m(v + \sigma d), \quad \sigma \in \{+1, -1\}, \quad (24)$$

so that $p_\sigma = W_{\sigma,q}$ in (2). A smooth external potential $U(q, t)$ is allowed: it adds $-U$ to the right side of (4)'s second equation and $\int \rho U dq$ to $\mathcal{H}$. The two sheet velocities $v \pm d$ are the forward and backward mean velocities of Nelson's stochastic mechanics with diffusion coefficient κ/m [Nelson1966; reading: abstract, as in the stochastic route]. The realization below replaces his Brownian motion by one sign that switches at a finite rate, as §8.23 of the recording note did for the Gaussian. The proofs are written derivations by Claude Fable (2026-10-01), checked against (146)–(149); they are unrefereed.

Proposition 6 (sheet field equations). On a smooth positive solution of (4) with potential U ,

$$\begin{aligned} v_t + v v_q &= -\frac{U_q}{m} + \frac{\kappa}{m} d_{qq} + d d_q, \\ d_t + v d_q + d v_q &= -\frac{\kappa}{m} v_{qq}. \end{aligned} \quad (25)$$

Proof. Put $L = \log \rho$, so $(\sqrt{\rho})_{qq}/\sqrt{\rho} = \frac{1}{2} L_{qq} + \frac{1}{4} L_q^2$. Differentiating (4)'s second equation in q and dividing by m gives $v_t = -v v_q - U_q/m + (\kappa^2/m^2)(L_{qqq} + L_q L_{qq})$, which is the first line because $(\kappa/m) d_{qq} = (\kappa^2/m^2) L_{qqq}$ and $d d_q = (\kappa^2/m^2) L_q L_{qq}$. Continuity gives $L_t = -v_q - v L_q$; its q -derivative times κ/m is the second line. $\square$

Proposition 7 (two-sheet realization: balance, force, energy). Let (Q_t, Σ_t) be a process on $\mathbb{R} \times \{\pm 1\}$ with velocity $\dot{Q} = v + \Sigma d$ between sign jumps, rate $\lambda_\sigma(q, t)$ from σ to $-\sigma$, and carried momentum $P_t = p_{\Sigma_t}(Q_t, t)$.

(i) *Balance.* The joint law is $\rho(q, t)/2$ for each sign at every time iff

$$\lambda_- - \lambda_+ = \frac{\kappa}{m} \frac{\rho_{qq}}{\rho}; \quad \lambda_\sigma = \frac{\kappa}{m} \left(-\sigma \frac{\rho_{qq}}{\rho} \right)_+ \text{ is the minimal nonnegative choice.} \quad (26)$$

(ii) *Force and impulse.* Between jumps P changes by

$$F_\sigma = -U_q + \kappa d_{qq} + 2m d d_q - \sigma \kappa v_{qq} = -\partial_q V_\sigma, \quad V_\sigma = U - \frac{\kappa^2}{m} \frac{\rho_{qq}}{\rho} + \sigma \frac{\kappa}{m} S_{qq}, \quad (27)$$

and at a jump $\sigma \rightarrow -\sigma$ by $\Delta p = -2\sigma m d = -2\sigma \kappa \partial_q \log \rho$. The symmetric part of the sheet potential is the rate difference in action units: $\frac{1}{2}(V_+ + V_-) - U = -\kappa(\lambda_- - \lambda_+)$.

(iii) *Energy.* With $K_\sigma = p_\sigma^2/(2m)$, pathwise $dK/dt = F_\Sigma \dot{Q}$ between jumps and $\Delta K = -2\Sigma m v d$ at a jump. In the mean, the κ -dependent part of the force and the jumps

have powers

$$\begin{aligned} W_F^\kappa &= \int \rho [\kappa v d_{qq} + 2m v d d_q - \kappa d v_{qq}] dq, \\ W_J &= \sum_\sigma \int \frac{\rho}{2} \lambda_\sigma (K_{-\sigma} - K_\sigma) dq = \kappa \int v d \rho_{qq} dq, \quad W_F^\kappa + W_J = 0, \end{aligned} \quad (28)$$

at every time, so that $\frac{d}{dt} E[K] = - \int \rho v U_q dq = d\mathcal{H}_{\text{kin}}/dt$. Individual trajectories exchange energy with the actuator and at jumps; the ensemble mean does not.

Proof. (i) The forward equation for the density n_σ of $(Q, \Sigma = \sigma)$ is $\partial_t n_\sigma + \partial_q (n_\sigma (v + \sigma d)) = -\lambda_\sigma n_\sigma + \lambda_{-\sigma} n_{-\sigma}$. With $n_\sigma = \rho/2$ and continuity, the left side is $\frac{\sigma}{2} \partial_q (\rho d) = \frac{\sigma \kappa}{2m} \rho_{qq}$ and the right side is $\frac{\rho}{2} (\lambda_{-\sigma} - \lambda_\sigma)$. The two signs give one equation and its negative. Given a difference, the nonnegative solutions are the displayed pair plus a common nonnegative rate. (ii) Along the sheet flow, $dP/dt = m [\partial_t + (v + \sigma d) \partial_q] (v + \sigma d) = m [(v_t + v v_q) + \sigma (d_t + v d_q + d v_q) + d d_q]$; insert (25). For the potential, $\kappa d_q + m d^2 = (\kappa^2/m)(L_{qq} + L_q^2) = (\kappa^2/m) \rho_{qq}/\rho$ and $\kappa v_q = (\kappa/m) S_{qq}$. The jump changes σd to $-\sigma d$. (iii) The pathwise statements restate (ii). For the mean, apply the generator to $g = K_\sigma(q, t)$: $\partial_t g + (v + \sigma d) \partial_q g = (v + \sigma d) F_\sigma$, and $K_{-\sigma} - K_\sigma = -2\sigma m v d$, so $W_J = \int \rho m v d (\lambda_- - \lambda_+) dq = \kappa \int v d \rho_{qq} dq$. The identity $W_F^\kappa + W_J = 0$ is verified directly: with $\rho d = (\kappa/m) \rho_q$, $\rho_{qq} = \rho (L_{qq} + L_q^2)$ and $\rho_{qqq} = \rho (L_{qqq} + 3L_q L_{qq} + L_q^3)$,

$$W_F^\kappa + W_J = \frac{\kappa^2}{m} \int [v \rho_{qqq} - \rho_q v_{qq}] dq = 0$$

after two integrations by parts with vanishing boundary terms. Since $E[K] = \int \rho \frac{m}{2} (v^2 + d^2) dq = \mathcal{H}_{\text{kin}}$, the last statement is the energy balance of (4) with potential. $\square$

On the Gaussian (145) one has $v = ax$, $d = -bx$, $d_{qq} = v_{qq} = 0$ and $\rho_{qq}/\rho = (z^2 - 1)/B$, so (26)–(28) reduce to (146), (148) and (149): $F_\sigma = 2mb^2x = 2\kappa^2x/(mB^2)$, $\lambda_+ = b(1 - z^2)_+$ and $W_J = -2mab^2B$. Three independent features are worth stating. The force is sign-dependent exactly through $-\sigma \kappa v_{qq}$, so it is sign-independent iff the mean velocity is affine; the Gaussian hid this. The force is a gradient of a potential that contains ρ_{qq}/ρ rather than the quantum potential $-(2\kappa^2/m)(\sqrt{\rho})_{qq}/\sqrt{\rho}$; the two differ by $-\frac{m}{2}d^2$, the sheet kinetic excess. And every trajectory of a state with zero mean current ($v \equiv 0$, for instance a trapped ground state) exchanges no energy at all, since both powers in (28) carry the factor v .

Proposition 8 (nonexplosion on the explicit orbit, all forward time). For (18) write $z = q/\sqrt{B}$, $s = \arctan u$, and $P(z, s) = 1 + 2\epsilon \sin(2s)H_2 + \epsilon^2 H_2^2 = |F|^2$, the factor of Proposition 4, so $\rho = \rho_G P/(1 + \epsilon^2)$. Then

$$\partial_z \log P = \frac{2\sqrt{2}\epsilon z (\sin 2s + \epsilon H_2)}{P}, \quad \partial_z \text{Arg } F = \frac{\sqrt{2}\epsilon z \cos 2s}{P}, \quad (29)$$

and in the clock s the sheet motion and rates read

$$\frac{dz}{ds} = -\sigma z + r_\sigma(z, s), \quad r_\sigma = 2\partial_z \text{Arg } F + \sigma \partial_z \log P, \quad \frac{\lambda_\sigma}{b} = \left(-\sigma B \frac{\rho_{qq}}{\rho} \right)_+ \leq c_\epsilon (1 + z^2), \quad (30)$$

with $|r_\sigma| \leq r_\epsilon$ and c_ϵ finite constants depending only on ϵ ; explicitly $|\partial_z \log P| \leq \max\{4\sqrt{2}\epsilon/(1 - \epsilon/\sqrt{2}), 8/3\}$. Hence $|z(s)| \leq (|z_0| + \frac{\pi}{2}r_\epsilon)e^{\pi/2}$ for all $t \geq 0$, the hazard integrated over the entire forward half-line is at most $\frac{\pi}{2}c_\epsilon(1 + \sup z^2) < \infty$ given Z_0 , the sign jumps are almost surely finitely many on $[0, \infty)$, and the process of Proposition 7 exists for all forward time with law (18). The orbit is even in q , so $v(0, t) = d(0, t) = 0$: the origin is invariant for both sheets and no trajectory crosses it.

Proof. With $H_2' = \sqrt{2}z$, $P_z = 2\epsilon H_2'(\sin 2s + \epsilon H_2)$ gives the first formula of (29). For the phase, $\partial_z \text{Arg } F = (\text{Re } F \partial_z \text{Im } F - \text{Im } F \partial_z \text{Re } F)/P$ with $\text{Re } F = 1 + \epsilon \sin(2s)H_2$, $\text{Im } F =$

$\epsilon \cos(2s)H_2$; the products of H_2H_2' cancel and leave $\epsilon \cos(2s)H_2'/P$. For the bound, note $P = (\sin 2s + \epsilon H_2)^2 + \cos^2 2s$, so $|\sin 2s + \epsilon H_2| \leq \sqrt{P}$ and $|\partial_z \log P| \leq 2\sqrt{2}\epsilon|z|/\sqrt{P}$. For $|z| \leq 2$ use $P \geq (1 - \epsilon/\sqrt{2})^2$ from (18). For $|z| \geq 2$, $H_2 \geq 0$ and $\sin 2s \geq 0$ give $P \geq \epsilon^2 H_2^2$ with $H_2 \geq 3z^2/(4\sqrt{2})$, so $|\partial_z \log P| \leq 16/(3|z|) \leq 8/3$. The same two regions bound $\partial_z \text{Arg } F$ and $\partial_z^2 \log P = P_{zz}/P - (P_z/P)^2$, where $P_{zz} = 2\sqrt{2}\epsilon(\sin 2s + \epsilon H_2 + \sqrt{2}\epsilon z^2)$. Now $S_q = \kappa uq/B + 2\kappa B^{-1/2}\partial_z \text{Arg } F$ and $\partial_q \log \rho = B^{-1/2}(-z + \partial_z \log P)$, so with $a = ub$, $v = \sqrt{B}(az + 2b\partial_z \text{Arg } F)$ and $d = \sqrt{B}b(-z + \partial_z \log P)$. Then $\dot{z} = (v + \sigma d)/\sqrt{B} - az = b[-\sigma z + 2\partial_z \text{Arg } F + \sigma \partial_z \log P]$, and $ds = b dt$ gives (30). Also $B\rho_{qq}/\rho = (-1 + \partial_z^2 \log P) + (-z + \partial_z \log P)^2$, which is bounded by $c_\epsilon(1 + z^2)$. Grönwall on $(|z|)' \leq |z| + r_\epsilon$ gives the growth bound; the clock runs only to $s = \pi/2$ over the whole physical half-line. Conditional on Z_0 , successive hazard clocks are dominated by one Poisson clock of finite rate, as in the proof of Proposition 21 of the recording note, which proves existence and nonexplosion; uniqueness of the forward solution with locally bounded rates identifies the law as $\rho/2$ per sign. Evenness of (18) in q gives the invariance of the origin. $\square$

Two limits of this realization are explicit. It needs $\rho > 0$: at a node, d and the rates diverge, and (18) is positive only forward in time. And it supplies no crossing, exactly as (145)–(148) did: packet crossing or relative-phase reunion cannot be tested on an even orbit. The posterior states (22) with the quadratic copy have $d \sim -(2\kappa/mC\ell^2)q^3$ at $t = 0$, so the $-$ sheet moves outward with cubic speed and would escape in finite time; its rate $\lambda_- \sim (m/\kappa)(2\kappa/mC\ell^2)^2 q^6$ has infinite integrated hazard before escape, so a switch to the inward $+$ sheet occurs almost surely first. Nonexplosion over an interval for these tails needs the evolved fields and is not claimed here.

7. The controller is the field pair

The realization needs four controller functions, $F_\pm(q, t)$ and $\lambda_\pm(q, t)$. Proposition 22 showed that they cannot be a preparation-independent law on (t, q, p, σ) . The following identification says exactly what they are.

Proposition 9 (controller identification). Work in the class of smooth positive ρ with $\rho, \rho_q \rightarrow 0$ at infinity and known U . (a) At one time, the rate difference determines ρ : with $W = (m/\kappa)(\lambda_- - \lambda_+)$, two positive decaying solutions of $\rho_{qq} = W\rho$ have a constant Wronskian that tends to zero, hence are proportional, and normalization fixes the factor. (b) Over any open time interval, the rates determine $\rho(\cdot, t)$, hence ρ_t , hence by continuity and decay $\rho v = -\int_{-\infty}^q \rho_t dq'$; so they determine v and S up to a constant. The forces are then fixed by (27) and add no data. (c) The forces alone also determine the field pair over an interval: $F_+ - F_- = -2\kappa v_{qq}$ gives v modulo affine functions; $\frac{1}{2}(F_+ + F_-) + U_q = (\kappa^2/m)\partial_q(\rho_{qq}/\rho)$ gives ρ_{qq}/ρ up to a constant c , and two positive decaying solutions of $\rho'' = (W_0 + c_i)\rho$ with $c_1 \neq c_2$ would have Wronskian increment $(c_2 - c_1) \int \rho_1 \rho_2 dq \neq 0$ between $-\infty$ and $+\infty$, contradicting decay; continuity then fixes the affine part of v . (d) At $\kappa = 0$ the rates vanish and $F_\sigma = -U_q$ for every preparation. The zero branch is exactly the branch on which the particle dynamics is autonomous.

Proof. (a) If $\rho_1, \rho_2 > 0$ solve $\rho'' = W\rho$ then $w = \rho_1\rho_2' - \rho_1'\rho_2$ has $w' = 0$, and $w \rightarrow 0$ at infinity by decay, so $(\rho_2/\rho_1)' = w/\rho_1^2 = 0$. (b) Continuity $\rho_t = -\partial_q(\rho v)$ with $\rho v \rightarrow 0$ at $-\infty$. (c) The displayed combinations follow from (27); the Wronskian computation is $w' = \rho_1\rho_2'' - \rho_1''\rho_2 = (c_2 - c_1)\rho_1\rho_2$. (d) Insert $d = 0$ in (26)–(27). $\square$

So either half of the controller is informationally the field pair. An apparatus implementing Proposition 7 stores (ρ, S) , or something from which (ρ, S) is computed, and updates it by (4). That is the pilot-wave structure: the field is a dynamical variable of the single system, carried alongside the particle. The known finite closed realization of the same law is the interacting-ensemble one, in which finitely many classical copies interact through a force built from the Fisher information of their empirical density and the field is recovered only as the copy number grows [HallDeckertWiseman2014; reading: metadata via Crossref and arXiv abstract]. There the controller datum is the configuration of the other copies and κ is the interworld coupling constant. In both realizations κ is a coupling that nothing in the construction fixes.

Corollary 10 (no finite controller survives one nonlinear record). Let $\mathcal{C}$ be a finite-dimensional family of field pairs containing (18), for instance Gaussian-polynomial states of bounded degree. A controller whose state ranges over $\mathcal{C}$ cannot realize the sheet dynamics after one quadratic coordinate copy of (18) with record r : the post-record pair (ρ_r, S) of Proposition 5 lies outside $\mathcal{C}$, and by Proposition 9 the controller functions on any interval after the copy determine that pair. The controller must therefore carry r ; after n copies it carries $(r_1, \dots, r_n)$, read or unread.

Proposition 11 (two unread widths, with a coordinate witness). Let a label $J \in \{1, 2\}$ with weights $\frac{1}{2}$ select preparations $N(0, A_j)$, $S = 0$, $A_1 \neq A_2$, and let J be unread. (a) *Branch realization.* Two controllers (145)–(148), one per A_j , give the mixture process; the sign stays fair given (Q, J) , the mean kinetic energy is $\frac{\kappa^2}{4m}(A_1^{-1} + A_2^{-1})$ for all time, and

$$EQ_t^2 = \frac{A_1 + A_2}{2} + \frac{\kappa^2 t^2}{2m^2} \left(\frac{1}{A_1} + \frac{1}{A_2} \right). \quad (31)$$

(b) *Single marginal-score controller.* Starting one controller from $\bar{\rho} = \frac{1}{2}(\rho_1 + \rho_2)$ and $\bar{S} = 0$ gives kinetic energy $\frac{\kappa^2}{2m}I(\bar{\rho})$ and, by the free virial identity $\frac{d^2}{dt^2}EQ^2 = 4\mathcal{H}_{\text{kin}}/m$, $EQ_t^2 = \frac{1}{2}(A_1 + A_2) + \kappa^2 I(\bar{\rho})t^2/m^2$. The two differ by

$$\frac{\kappa^2 t^2}{m^2} \Delta, \quad \Delta = \frac{1}{2} \left(\frac{1}{A_1} + \frac{1}{A_2} \right) - I(\bar{\rho}) = \left(\frac{1}{A_1} - \frac{1}{A_2} \right)^2 \int \frac{q^2 \rho_1 \rho_2}{4\bar{\rho}} dq > 0, \quad (32)$$

which is (119) for this mixture. The single controller loses energy $\kappa^2 \Delta/(2m)$ and underpredicts the variance by $\kappa^2 \Delta t^2/m^2$ at every $t > 0$: a difference visible in coordinate records alone.

Proof. (a) Each branch is Proposition 21 with its own A_j ; the energies and variances are (143)'s, averaged. (b) The virial identity is the time derivative of (19)'s structure for any free solution of (4): $\frac{d}{dt} \int \rho q^2 = 2 \int \rho q v$ and $\frac{d^2}{dt^2} \int \rho q^2 = 2 \int \rho (v^2 + d^2) dq$ by (25) and two integrations by parts, which is $4\mathcal{H}_{\text{kin}}/m$. For Δ , the conditional label law is $\pi_j(q) = \rho_j/(2\bar{\rho})$ and the branch scores are $-q/A_j$, so $\text{Var}(\sigma_J(q) | q) = q^2 \pi_1 \pi_2 (A_1^{-1} - A_2^{-1})^2$; insert in (119). $\square$

Within this model, the rule against replacing a discarded label by a fresh marginal score is therefore a theorem with a coordinate witness, and closure under unread records holds only for branch controllers. Combined with Corollary 10: the controller of a closed realization is the complete record history together with the initial field pair.

Proposition 12 (closure within the theory, at every κ). Coordinate shears $f(q)\pi$ preserve the class (2)/(136) with its fair sign (Proposition 20 of the recording note), and the sheet dynamics of Proposition 7, applied to body and pointers alike with their own potentials, preserves it by construction wherever $\rho > 0$. (For several configuration coordinates with one shared sign, Propositions 6–7 hold with ∇ and the mass-weighted Laplacian $\kappa \sum_a \partial_a^2 \rho / (m_a \rho)$ in place of ∂_q and $(\kappa/m)\rho_{qq}/\rho$: because v and d are gradients, the cross terms $\sigma[(d \cdot \nabla)v - (\nabla v)d]$ in the force cancel by symmetry of $\partial_a \partial_b S$.) Hence every finite composition of coordinate copies, sheet evolution and conditioning on coordinate records stays in the class, and the floor (138) holds for every complete coordinate record. The escape (140) reads a canonical momentum directly; within the theory a momentum is accessed only through coordinate copies after sheet evolution, which keeps the floor. The same closure holds trivially at $\kappa = 0$, where the sheets coincide and (138) reads $\det \geq 0$. Closure under composition and complete coordinate records therefore holds for every $\kappa \geq 0$ and selects no scale. Neither branch is unconditionally closed: the $\kappa = 0$ single-sheet class leaves itself at caustics of the classical flow, and the $\kappa > 0$ realization needs $\rho > 0$ and is not continued through nodes here.

8. Assessment: a representation, not an apparatus

The proposed dynamical constraint realization exists and is exact: Propositions 7–8 give its balance, force, impulses, energy exchange and all-time nonexplosion on the positive non-Gaussian

orbit, and Proposition 12 gives its closure under coordinate copies. It is a stochastic pilot-wave representation of the supplied field law, with a switching sign in place of Nelson’s diffusion. Proposition 9 then answers the handout’s question: the extra state that controls the force and the rates is the field pair itself, in either half of the controller, and after records it is the complete record history. The construction simulates (4); it does not supply a physical rule that would make the particle dynamics autonomous, and Proposition 9(d) shows that autonomy is exactly the zero branch.

Consequently the route “close the apparatus and read off a positive scale” ends here. Every closed realization of the field law carries κ as a coupling, between particle and field or between copies, and closure is available at every $\kappa \geq 0$ (Proposition 12). What a positive scale requires is a premise that fails at $\kappa = 0$, and Proposition 9(d) states its physical content without quantum language: *the motion of an individual body depends on the statistical state of its preparation, not only on its own position, momentum and the material parts of the apparatus.* Newton’s inflexion Observation 8 records dependence on apparatus geometry, which admits a local-force countermodel (routes, Proposition 3); it is not evidence for dependence on the ensemble. The two live forms of the missing premise remain the terminal readout bound and Leibniz continuity on records; the present note adds that no further realization of the field law can replace them.

Retained from this unit for later use: the exact particle picture that is consistent with every time partition (Proposition 8), the explicit sheet potential and powers (27)–(28), and the coordinate witness (32) for the unread-label excess.

9. Consequence for STATE

The canonical Fisher structure follows from a specified restriction of ordinary classical sheet action; a fixed compact-transfer family fails even with Gaussian width; (18)–(20) complete a width/phase tangent to an exact positive non-Gaussian forward orbit; a nonlinear record generates a quartic tail outside the finite mode family with unread energy (23). The two-sheet dynamical realization on that orbit is now a theorem (Propositions 7–8), with closure under coordinate copies (Proposition 12) and the two-unread-width test passed by branch controllers and failed, with a coordinate witness, by any marginal-score controller (Proposition 11). Its controller is the field pair and the record history (Proposition 9, Corollary 10): the realization is a stochastic pilot-wave representation of the supplied law, and closure holds at every $\kappa \geq 0$. Decision: the realization route is closed as a source of positive action necessity; its results are kept as the partition-consistent particle picture. The next Newton lemma must attack the premise that fails at $\kappa = 0$, dependence of individual motion on the preparation’s statistical state, through the terminal readout bound or the record reading of Leibniz continuity, with positivity, universality and radiation calibration still separate. Reject a further realization of (4) offered as apparatus closure, a projected residual counted as exact closure, bridge phase used as population momentum without reservoir costs, or an unexcluded zero branch presented as positive action necessity.